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// Behavioral description of a 4-bit universal shift register
// Fig. 6.7 and Table 6.3
module Shift_Register_4_beh ( // V2001, 2005
    output reg [3: 0] A_par, // Register output
    input [3: 0] I_par, // Parallel input
    input s1, s0, // Select inputs
    MSB_in, LSB_in, // Serial inputs
    CLK, Clear_b // Clock and active-low clear
);
always @ (posedge CLK, negedge Clear_b) // V2001, 2005
    if (Clear_b == 0) A_par <= 4'b0000;
    else
        case ({s1, s0})
            2'b00: A_par <= A_par;
                // No change
            2'b01: A_par <= {MSB_in, A_par[3: 1]}; // Shift right
            2'b10: A_par <= {A_par[2: 0], LSB_in}; // Shift left
            2'b11: A_par <= I_par;
                // Parallel load of input
        endcase;
endmodule

```
